DYLAN ASHLEY

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EDUCATION

Ph.D. in Informatics Jan. 2021 – Present
Università della Svizzera italiana (Dalle Molle Institute for Artificial Intelligence Research)
Supervisor: Jürgen Schmidhuber
Focus: Agentic AI, Creative Systems, Reinforcement Learning, Robotics, Neural Networks

M.Sc. in Computing Science Sep. 2017 – Nov. 2020
University of Alberta (Alberta Machine Intelligence Institute)
Supervisor: Richard S. Sutton **(2024 Turing Award Recipient)**
GPA: 4.0 / 4.0

B.Sc. Honors in Computing Science Sep. 2013 – Jun. 2017
University of Alberta

WORK EXPERIENCE

Research Scientist Intern Jun. 2026 – Nov. 2026
Meta Platforms, Inc.

- Conducted research at the Burlingame office under the supervision of Changsheng Zhang and Yangyang Shi on Agentic AI.

Research Scientist Intern May 2025 – Nov. 2025
Meta Platforms, Inc.

- Conducted research at the Zürich office under the supervision of Gaël Le Lan on Agentic AI.

Visiting Researcher Jan. 2023 – Present
Center of Excellence in Generative AI, King Abdullah University of Science and Technology

- Conducting research with Prof. Eric Feron and Prof. Jürgen Schmidhuber on several topics related to deep learning and robotics.
- Co-lead the writing of two successful grants worth US$ 2’376’300 and US$ 850’000.

Chief Technology Officer May 2021 – Present
Perseverance Analytics Ltd.

- Founding member of an incorporated non-profit data science startup based in Alberta, Canada.
- Startup focuses on connecting communities to social supports to bridge the gap between the availability and accessibility of services.

Doctoral Research Assistant Jan. 2021 – Present

Università della Svizzera italiana

- Conducting research with Prof. Jürgen Schmidhuber on several deep learning topics.

Vice-President Academic May 2019 – Apr. 2020

Graduate Students' Association, University of Alberta

- Salaried official representative of over 7,900 graduate students in academic matters.
- Advocated for graduate student issues to the university and worked with the university to build a better learning environment for students.
- Delivered several significant advocacy victories, including better oversight for mentorship and a reduced increment in tuition during a budgetary crisis.
- Time commitment of approximately 30 hours a week for a one-year term.

Graduate Research Assistant Sep. 2017 – Aug. 2020

Reinforcement Learning and Artificial Intelligence Lab, University of Alberta

- Conducted research with Prof. Richard S. Sutton and others on several reinforcement learning topics.

Undergraduate Summer Research Assistant May 2015 – Aug. 2017

University of Alberta

- Won three separate competitive four-month NSERC Undergraduate Student Research Awards, the first working with first Prof. José Nelson Amaral, and the latter two working with Prof. Richard S. Sutton.

PEER-REVIEWED PUBLICATIONS

[1] Dai, Y.*, Wang, Y.*, **Ashley, D. R.**, & Schmidhuber, J. (2026). Efficient Morphology-Control Co-Design via Stackelberg Proximal Policy Optimization. *Proceedings of the 14th International Conference on Learning Representations.* `https://openreview.net/pdf?id=sJ0vOOkclw`

[2] Alhakami, M.*, **Ashley, D. R.***, Dunham, J.*, Dai, Y., Faccio, F., Feron, F., & Schmidhuber, J. (2025). Towards an Extremely Robust Baby Robot With Rich Interaction Ability for Advanced Machine Learning Algorithms. *Proceedings of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems.* `https://doi.org/10.1109/IROS60139.2025.11246188`

Oral Presentation. Previously presented as a late-breaking result at the 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

[3] Zhuge, M., Zhao, C., **Ashley, D. R.**, Wang, W., Khizbullin, D., Xiong, Y., Liu, Z., Chang, E., Krishnamoorthi, R., Tian, Y., Shi, Y., Chandra, V., & Schmidhuber, J. (2025). Agent-as-a-Judge: Evaluate Agents with Agents. *Proceedings of the 42nd International Conference on Machine Learning.* `https://openreview.net/pdf?id=Nn9POI9Ekt`

[4] Wang, Y., Wu, Q., **Ashley, D. R.**, Faccio, F., Li, W., Huang, C., & Schmidhuber, J. (2025). Scaling Value Iteration Networks to 5000 Layers for Extreme Long-Term Planning. *Proceedings of the 42nd International Conference on Machine Learning.* `https://openreview.net/pdf?id=YwltsnSUFQ`

Previously presented at the 2024 European Workshop on Reinforcement Learning (EWRL).

[5] Zhuge, M.*, Liu, H.*, Faccio, F.*, **Ashley, D. R.***, Csordas, R., Gopalakrishnan, A., Hamdi, A., Hammoud, H. A. A. K., Herrmann, V., Irie, K., Kirsch, L., Li, B., Li, G., Liu, S., Mai, J., Piekos, P., Ramesh, A., Schlag, I., Shi, W., Stanic, A., Wang, W., Wang, Y., Xu, M., Fan, D.-P., Ghanem, B., & Schmidhuber, J. (2025). Mindstorms in natural language-based societies of mind. *Computational Visual Media* **(IF 17.3)**, *11*(1), 29–81. `https://doi.org/10.26599/CVM.2025.9450460`

Previously presented at the NeurIPS 2023 Workshop on Robustness of Zero/Few-Shot Learning in Foundation Models **(Best-paper Award)**.

[6] **Ashley, D. R.***, Herrmann, V.*, Friggstad, Z., & Schmidhuber, J. (2025). On narrative information and the distillation of stories. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **(IF 20.8)**, *47*(2), 697–707. `https://doi.org/10.1109/TPAMI.2024.3480702`

Previously presented at the NeurIPS 2023 Workshop on Machine Learning for Creativity and Design and at the NeurIPS 2022 Workshop Information-Theoretic Principles in Cognitive Systems.

[7] Štrupl, M., Faccio, F., **Ashley, D. R.**, Srivastava, R. K., & Schmidhuber, J. (2022). Reward-Weighted Regression Converges to a Global Optimum. *Proceedings of the 36th AAAI Conference on Artificial Intelligence,* 8361–8369. `https://doi.org/10.1609/aaai.v36i8.20811`

[8] **Ashley, D. R.** (2020). *Understanding Forgetting in Artificial Neural Networks* [Master's thesis, University of Alberta]. University of Alberta Education and Research Archive. `https://doi.org/10.7939/r3-6zvv-5z64`

[9] **Ashley, D. R.***, Chockalingam, V.*, Kuzma, B.*, & Bulitko, V. (2019). Learning to Select Mates in Evolving Non-playable Characters. *Proceedings of the 2019 IEEE Conference on Games,* 1–8. `https://doi.org/10.1109/CIG.2019.8848114`

[10] **Ashley, D. R.***, Chockalingam, V.*, Kuzma, B.*, & Bulitko, V. (2019). Learning to Select Mates in Artificial Life. *Proceedings of the Genetic and Evolutionary Computation Conference Companion,* 103–104. `https://doi.org/10.1145/3319619.3322060`

[11] Sherstan, C., **Ashley, D. R.***, Bennett, B.*, Young, K., White, A., White, M., & Sutton, R. S. (2018). Comparing Direct and Indirect Temporal-Difference Methods for Estimating the Variance of the Return. *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence,* 63–72. `http://auai.org/uai2018/proceedings/papers/35.pdf`

Oral Presentation.

[12] Amaral, J. N., Borin, E., **Ashley, D. R.**, Benedicto, C., Colp, E., Hoffmam, J. H. S., Karpoff, M., Ochoa, E., Redshaw, M., & Rodrigues, R. E. (2018). The Alberta Workloads for the SPEC CPU 2017 Benchmark Suite. *Proceedings of the 2018 IEEE International Symposium on Performance Analysis of Systems and Software,* 159–168. https://doi.org/10.1109/ISPASS.2018.00029

* equal contribution

PREPRINTS AND TECHNICAL REPORTS

[13] **Ashley, D. R.**, Le Lan, G., Zhao, C., Dhingra, N., Cai, Z., Chang, E., Zhuge, M., Shi, Y., Chandra, V., & Schmidhuber, J. (2026). RPRA: Predicting an LLM-Judge for Efficient but Performant Inference. https://arxiv.org/abs/2604.12634

Under review. Previously presented at the ICLR 2026 Workshop on Multi-Agent Learning and Its Opportunities in the Era of Generative AI (RLDM) and the ICLR 2026 Workshop on Agentic AI in the Wild: From Hallucinations to Reliable Autonomy.

[14] **Ashley, D. R.***, Przepióra, J.*, Chen, Y.*, Abualsaud, A., Yesmagambet, N., Park, S., Feron, E., & Schmidhuber, J. (2026). RACAS: Controlling Diverse Robots With a Single Agentic System. https://arxiv.org/abs/2603.05621

Under review.

[15] Štrupl, M., Szehr, O., Faccio, F., **Ashley, D. R.**, Srivastava, R. K., & Schmidhuber, J. (2025). *On the Convergence and Stability of Upside-Down Reinforcement Learning, Goal-Conditioned Supervised Learning, and Online Decision Transformers.* https://arxiv.org/abs/2502.05672

Under review.

[16] Di Ventura, J., **Ashley, D. R.**, Herrmann, V., Faccio, F., & Schmidhuber, J. (2025). *Upside Down Reinforcement Learning with Policy Generators.* https://arxiv.org/abs/2501.16288

Presented at the 2025 Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM) and the 2025 European Workshop on Reinforcement Learning (EWRL).

[17] Wang, W., Alyahya, H. A., **Ashley, D. R.**, Serikov, O., Khizbullin, D., Faccio, F., & Schmidhuber, J. (2024). *How to Correctly do Semantic Backpropagation on Language-based Agentic Systems.* http://arxiv.org/abs/2412.03624

Under review.

[18] Herrmann, V.*, **Ashley, D. R.***, & Schmidhuber, J. (2024). *Automatic Album Sequencing.* https://arxiv.org/abs/2411.07772

Presented as a late-breaking demo at the 2024 Conference of the International Society for Music Information Retrieval (ISMIR).

[19] Stanic, A.*, **Ashley, D.**, Serikov, O., Kirsch, L., Faccio, F., Schmidhuber, J., Hofmann, T., & Schlag, I.* (2023). *The languini kitchen: Enabling language modelling research at different scales of compute.* https://arxiv.org/abs/2309.11197

[20] Štrupl, M., Faccio, F., **Ashley, D. R.**, Schmidhuber, J., & Srivastava, R. K. (2022). *Upside-Down Reinforcement Learning Can Diverge in Stochastic Environments With Episodic Resets.* `https://arxiv.org/abs/2205.06595`

Presented at the 2022 Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM).

[21] **Ashley, D. R.**, Arulkumaran, K., Schmidhuber, J., & Srivastava, R. K. (2022). *Learning Relative Return Policies With Upside-Down Reinforcement Learning.* `https://arxiv.org/abs/2202.12742`

Presented at the 2022 Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM).

[22] Arulkumaran, K., **Ashley, D. R.**, Schmidhuber, J., & Srivastava, R. K. (2022). *All You Need Is Supervised Learning: From Imitation Learning to Meta-RL With Upside Down RL.* `https://arxiv.org/abs/2202.11960`

Presented at the 2022 Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM).

[23] **Ashley, D. R.**, Ghiassian, S., & Sutton, R. S. (2021). *Does the Adam optimizer exacerbate catastrophic forgetting?* `https://arxiv.org/abs/2102.07686`

[24] Ma, C., **Ashley, D. R.**, Wen, J., & Bengio, Y. (2020). *Universal successor features for transfer reinforcement learning.* `https://arxiv.org/abs/2001.04025`

* equal contribution

AWARDS

Best-Paper Award, NeurIPS 2023 Ro-FoMo Workshop	2023
Rising Star in AI Award, King Abdullah University of Science and Technology	2022
Queen Elizabeth II Graduate Scholarship, University of Alberta (C$10,800)	2018
CGS-M, Natural Science and Engineering Research Council of Canada (C$17,500)	2017
Walter H. Johns Graduate Fellowship, University of Alberta (C$5,800)	2017
Science Graduate Scholarship, University of Alberta (C$2,000)	2017
Kao Family Eisenco Scholarship, University of Alberta (C$1,200)	2016
Jason Lang Scholarship, University of Alberta (C$1,000)	2015
Suncor Energy Scholarship, Suncor Energy (C$1,800)	2015
Jason Lang Scholarship, University of Alberta (C$1,000)	2014
Suncor Energy Scholarship, Suncor Energy (C$1,800)	2014

INVITED TALKS

International Conference on Intelligent Robots and Systems, Hangzhou, CN	2025
NeurIPS 2023 Ro-FoMo Workshop, New Orleans, US	2023
KAUST Rising Stars in AI Symposium, Thuwal, SA	2022
Alberta Machine Intelligence Institute Tea Time Talks, Virtual	2021
Canadian Association for Graduate Studies Conference, Halifax, CA	2019
IEEE Conference on Games, London, GB	2019

Chinese Graduate Students Association Workshop, Edmonton, CA 2019
Kindred, Toronto, CA 2018

SERVICE

Reviewer, Conference on Language Modeling 2026
Reviewer, International Conference on Intelligent Robots and Systems 2026
Reviewer, International Conference on Machine Learning 2026
Reviewer, ICLR Workshop on Multi-Agent Learning and Generative AI 2026
Reviewer, ICLR Workshop on Agentic AI in the Wild: Reliable Autonomy 2026
Reviewer, International Conference on Learning Representations 2026
Reviewer, AAAI Conference on Artificial Intelligence 2026
Reviewer, ICLR Workshop on World Models 2025
Reviewer, International Conference on Learning Representations 2025
Reviewer, European Workshop on Reinforcement Learning 2024
Reviewer, NeurIPS Workshop on Aligning RL Experimentalists and Theorists 2024
Reviewer, International Conference on Artificial Neural Networks 2024
Reviewer, Machine Learning 2024
Program Committee Member, European Workshop on Reinforcement Learning 2023
Reviewer, European Workshop on Reinforcement Learning 2023
Reviewer, ICML Workshop on Interactive Learning with Implicit Human Feedback 2023
Reviewer, Machine Learning 2023
Faculty of Informatics Council, Università della Svizzera italiana 2022 – 2023
Reviewer, NeurIPS Workshop on Information-Theoretic Principles in Cognitive Systems 2022
Reviewer, NeurIPS Workshop on Reinforcement Learning for Real Life Workshop 2022
Reviewer, European Workshop on Reinforcement Learning 2022
Mentoring Award Adjudication Panel, University of Alberta 2020
Equity, Diversity, and Inclusion Council, Alberta Machine Intelligence Institute 2019 – 2020
Volunteer, Campus Food Bank 2019
Faculty of Graduate Studies and Research Council, University of Alberta 2018 – 2019
Board Member, Graduate Students' Association 2018 – 2019
Nominating Committee, Graduate Students' Association 2018 – 2019
Council Member, Graduate Students' Association 2018 – 2020

SUPERVISING

Jacopo di Ventura, M.Sc. Student 2023 – 2024
James Jewitt, High School Intern 2015

TEACHING

Teaching Assistant, Machine Learning	2024
Teaching Assistant, Machine Learning	2023
Teaching Assistant, Machine Learning	2022
Teaching Assistant, Algorithms & Data Structures	2022
Teaching Assistant, Machine Learning	2021
Teaching Assistant, Introduction to File and Database Management	2015

PROFESSIONAL MEMBERSHIP

Association for Computing Machinery	Since 2014
Association for the Advancement of Artificial Intelligence	Since 2018
Institute of Electrical and Electronics Engineers	Since 2019

NATURAL LANGUAGES

English: Native Speaker	**French:** Moderate Fluency
Italian: Some Knowledge	**Mandarin:** Some Knowledge

PROGRAMMING LANGUAGES

Python: Expert	**Bash:** Strong	**C/C++:** Strong	**LaTeX:** Strong
Java: Intermediate	**SQL:** Intermediate	**HTML:** Basic	**Lisp:** Basic
Prolog: Basic			

REFERENCES AVAILABLE ON REQUEST